

Phase 1

Vehicle insurance Database creation

Introduction:

This project develops a comprehensive relational database management system for vehicle insurance policy administration. The system is designed to efficiently manage and maintain structured data related to policyholders, vehicles, insurance staff, coverage plans, transactions, and claims processing.

It incorporates advanced RDBMS concepts such as entity-relationship modeling, normalization, primary and foreign key constraints, referential integrity, joins, nested queries, aggregate functions, views, stored procedures, and triggers. These features ensure data consistency, minimize redundancy, and enforce business rules within the system.

The implementation demonstrates how a well-structured relational database can optimize insurance management operations, enhance data security, and support efficient query processing for decision-making.

Objective:

1. To create a database system for managing vehicle insurance details.
2. To store and manage customer, vehicle, policy, staff, payment, and claim information.
3. To use primary keys and foreign keys to connect related tables.
4. To reduce data duplication and maintain data accuracy.
5. To perform operations like joins, subqueries, views, and triggers for efficient data handling.
6. To apply RDBMS concepts in a practical, real-time application.

Database creation:

Query:

Create database project:

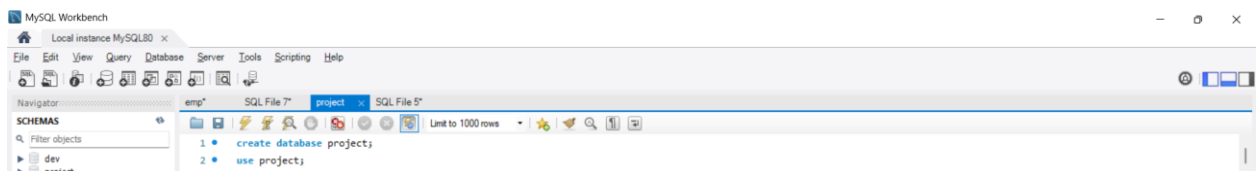


Table1:

Customers:

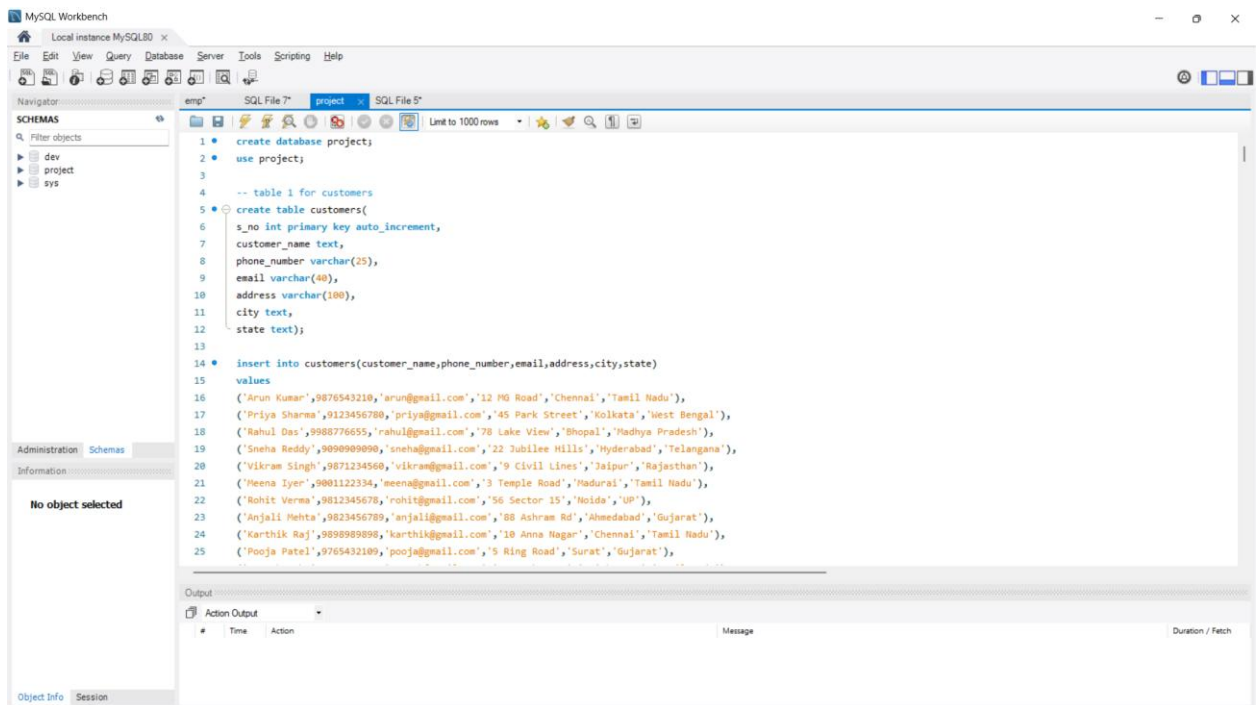


Table 2:

Vehicles

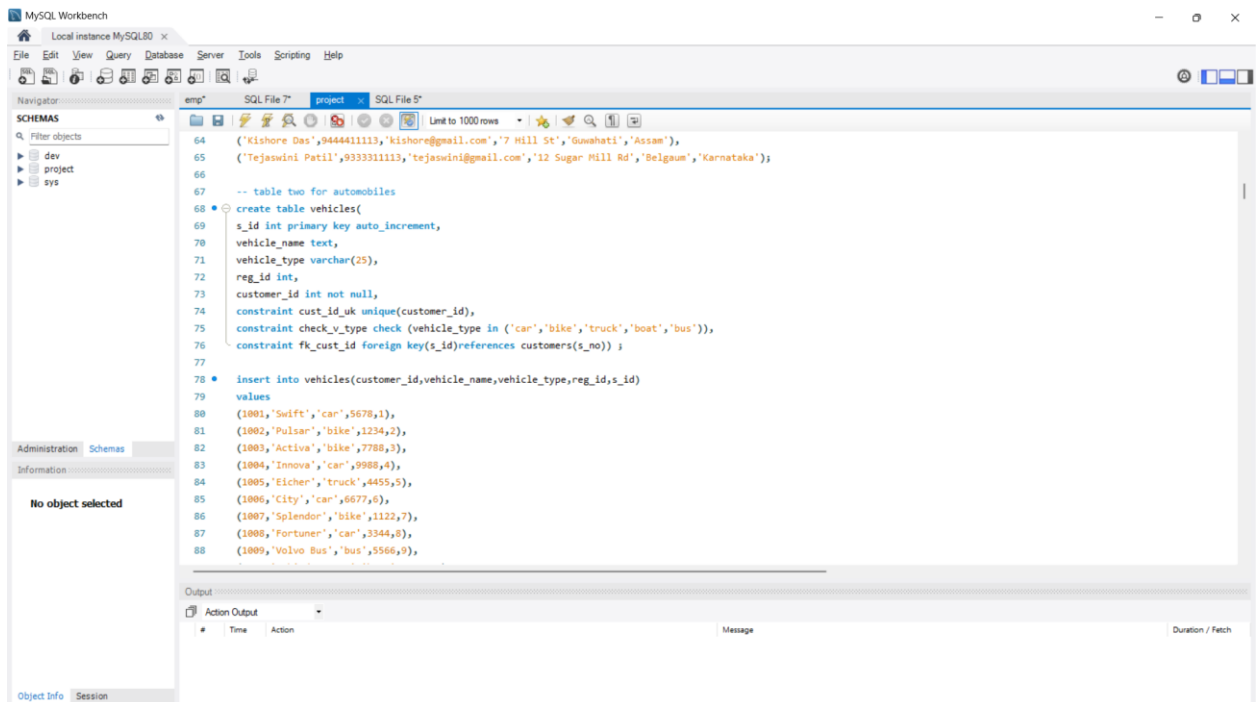


Table3:

policy

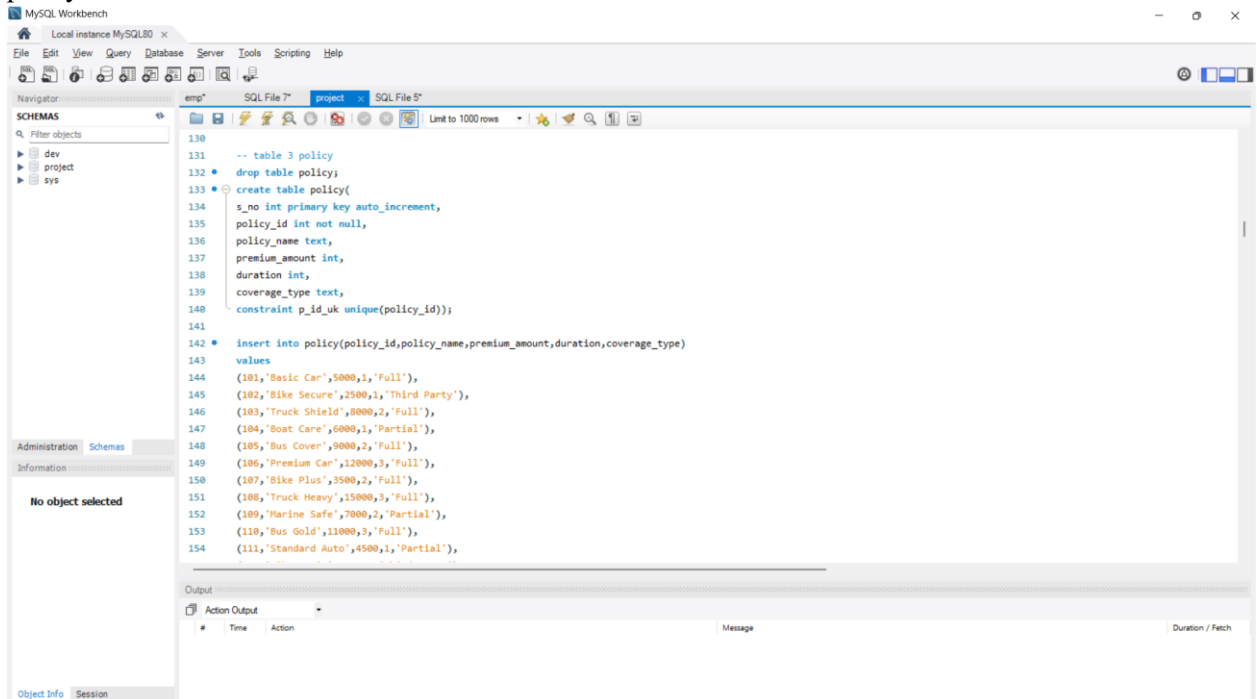


Table 4:

insurance

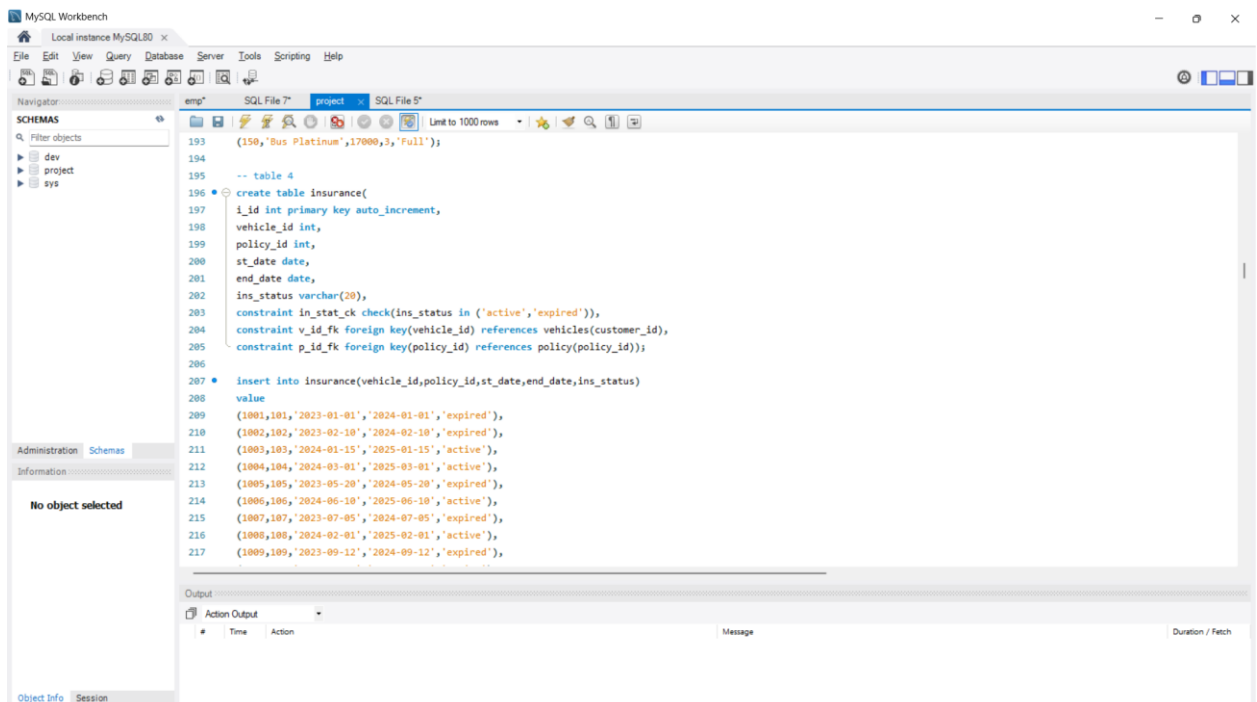


Table5:

claims

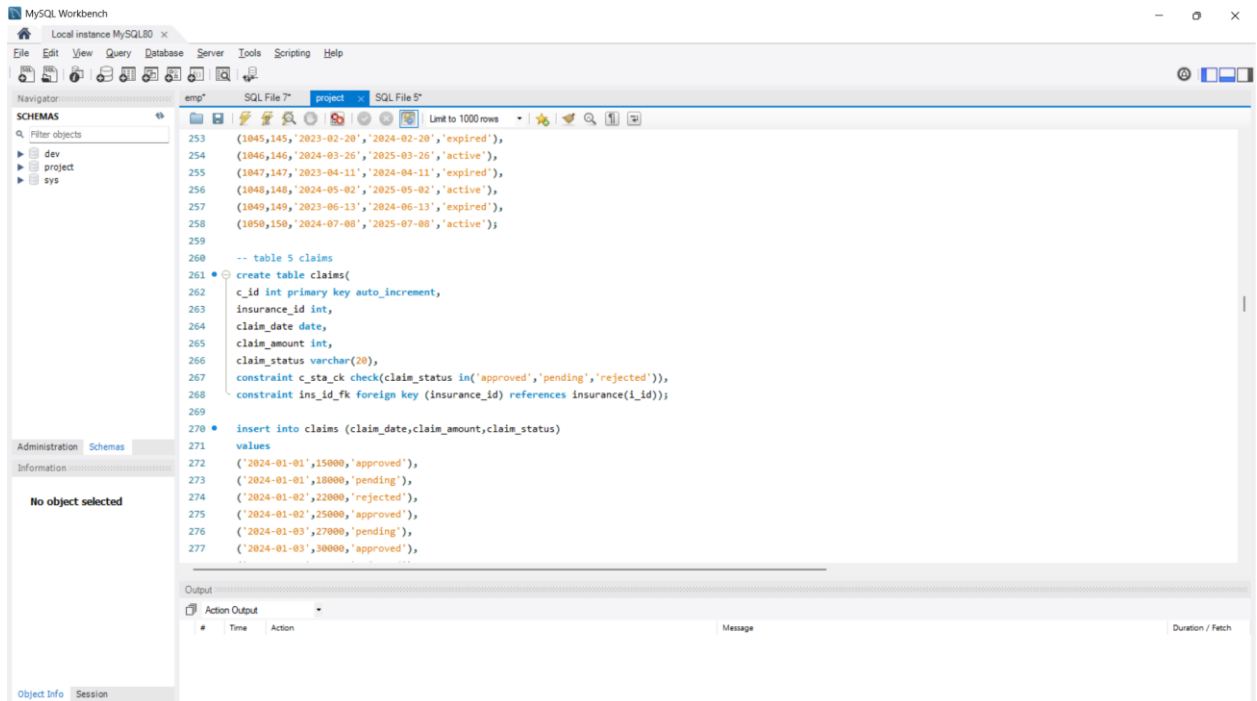


Table 6:

payment

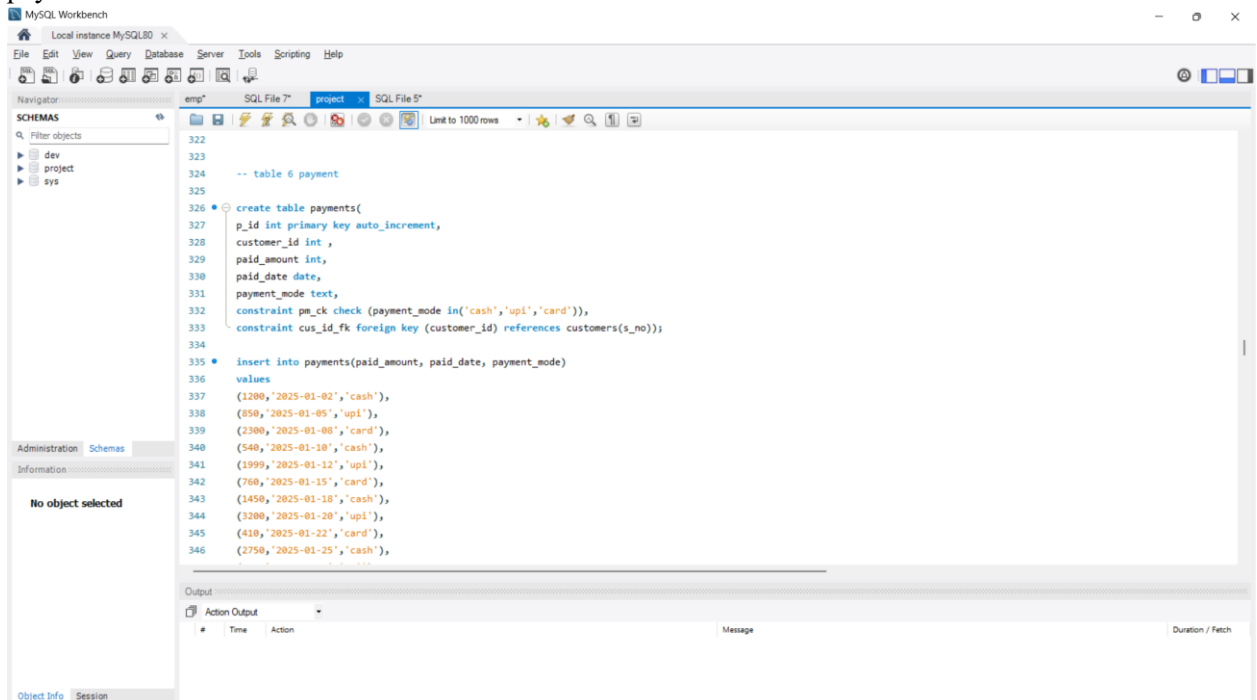


Table 7:

agents

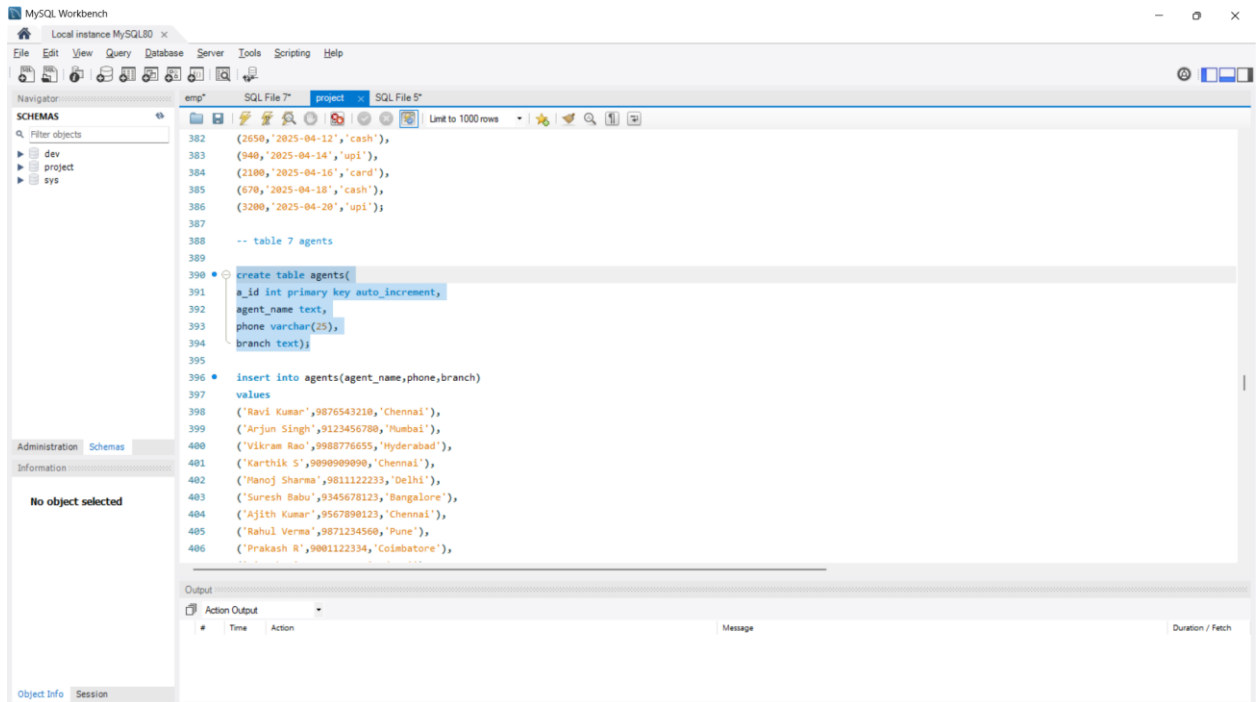


Table 8:
relations

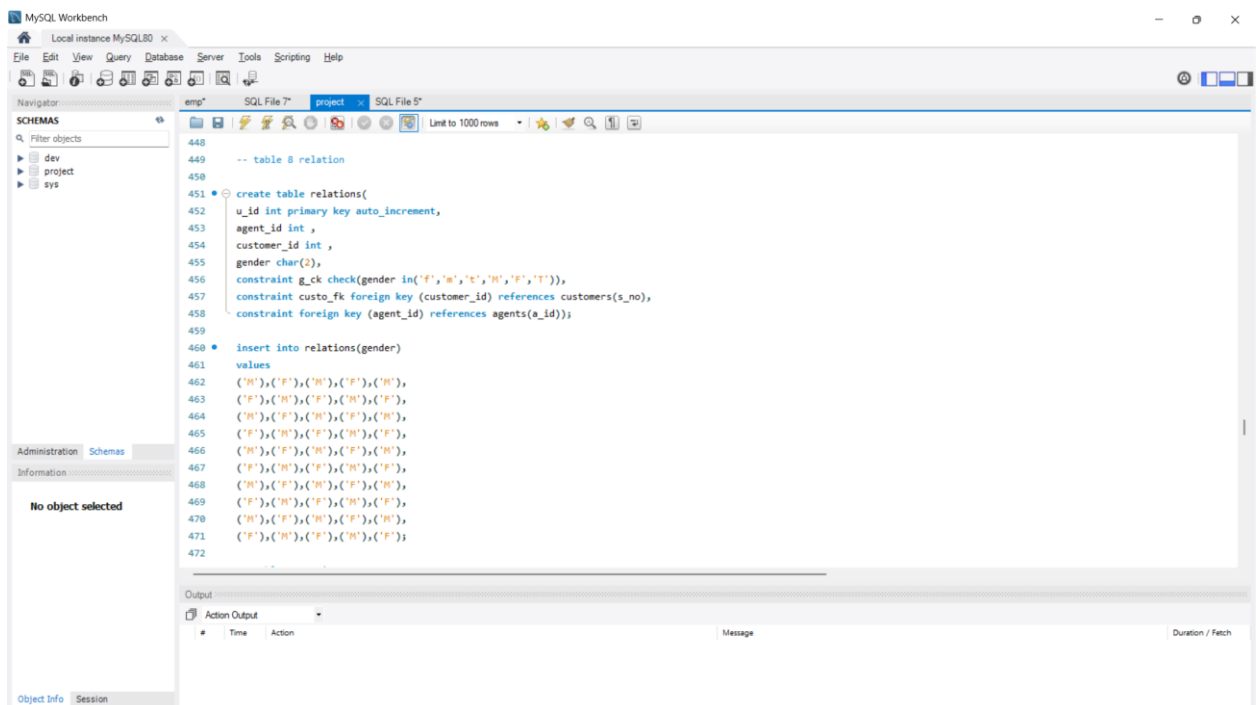


Table9:

services

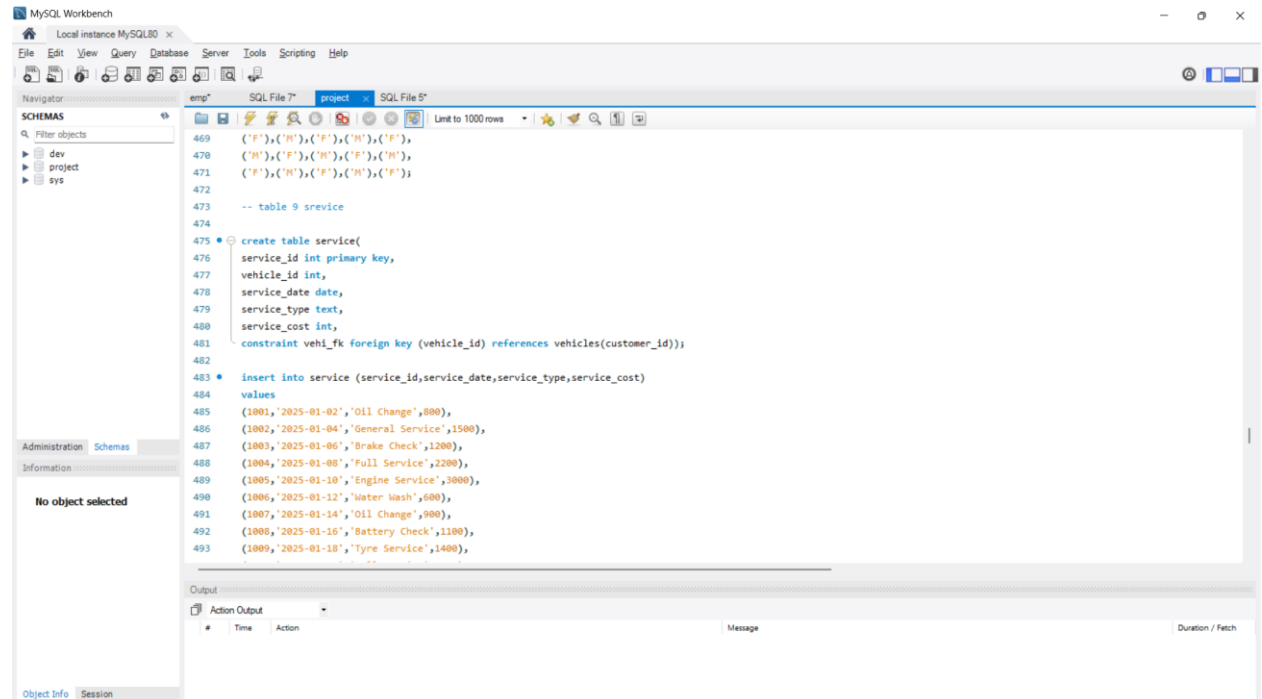
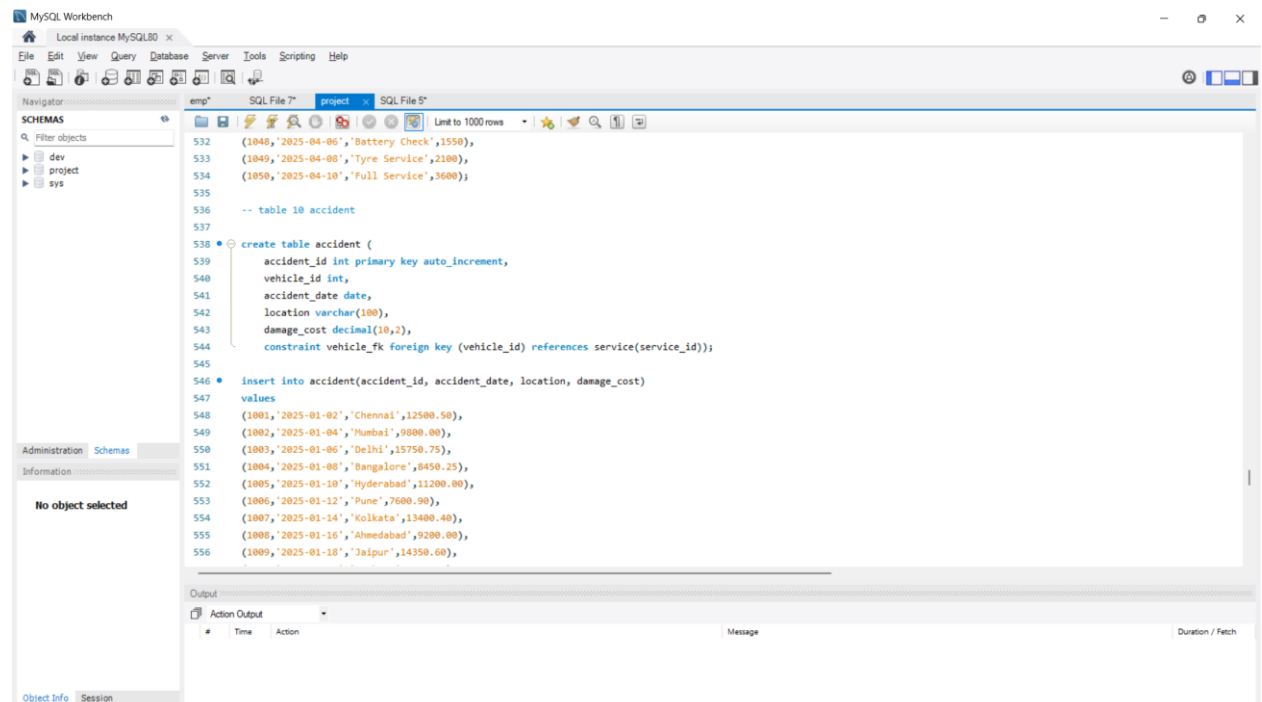


Table10:

accidents



Insert table values:

Query

Table 1 customers

insert into customers(customer_name,phone_number,email,address,city,state)

values

('Arun Kumar',9876543210,'arun@gmail.com','12 MG Road','Chennai','Tamil Nadu'),
('Priya Sharma',9123456780,'priya@gmail.com','45 Park Street','Kolkata','West Bengal'),
('Rahul Das',9988776655,'rahul@gmail.com','78 Lake View','Bhopal','Madhya Pradesh'),
('Sneha Reddy',9090909090,'sneha@gmail.com','22 Jubilee Hills','Hyderabad','Telangana'),
('Vikram Singh',9871234560,'vikram@gmail.com','9 Civil Lines','Jaipur','Rajasthan'),
('Meena Iyer',9001122334,'meena@gmail.com','3 Temple Road','Madurai','Tamil Nadu'),
('Rohit Verma',9812345678,'rohit@gmail.com','56 Sector 15','Noida','UP'),
('Anjali Mehta',9823456789,'anjali@gmail.com','88 Ashram Rd','Ahmedabad','Gujarat'),
('Karthik Raj',9898989898,'karthik@gmail.com','10 Anna Nagar','Chennai','Tamil Nadu'),
('Pooja Patel',9765432109,'pooja@gmail.com','5 Ring Road','Surat','Gujarat'),
('Suresh Babu',9345678901,'suresh@gmail.com','77 Market St','Coimbatore','Tamil Nadu'),
('Neha Kapoor',9012345678,'neha@gmail.com','2 Film City','Mumbai','Maharashtra'),
('Deepak Yadav',9123987654,'deepak@gmail.com','44 Bus Stand Rd','Patna','Bihar'),
('Kiran Nair',9891234567,'kiran@gmail.com','16 Beach Rd','Kochi','Kerala'),
('Amit Shah',9988123456,'amit@gmail.com','1 River View','Vadodara','Gujarat'),
('Lakshmi Devi',9876501234,'lakshmi@gmail.com','90 Flower Rd','Trichy','Tamil Nadu'),
('Mohit Jain',9811198765,'mohit@gmail.com','34 Main Bazaar','Delhi','Delhi'),
('Divya Shetty',9765001122,'divya@gmail.com','7 Hill Road','Mangalore','Karnataka'),
('Ganesh Pawar',9700011223,'ganesh@gmail.com','19 Station Rd','Pune','Maharashtra'),
('Nisha Gupta',9696969696,'nisha@gmail.com','4 Colony Rd','Lucknow','UP'),
('Ajay Kumar',9555512345,'ajay@gmail.com','55 Gandhi Rd','Kanpur','UP'),
('Rekha Pillai',9444412345,'rekha@gmail.com','21 Lake Rd','Trivandrum','Kerala'),
('Harish Rao',9333312345,'harish@gmail.com','9 IT Park','Hyderabad','Telangana'),
('Komal Arora',9222212345,'komal@gmail.com','12 Model Town','Ludhiana','Punjab'),
('Sunil Joshi',9111112345,'sunil@gmail.com','89 Fort Area','Mumbai','Maharashtra'),
('Preeti Saxena',9000012345,'preeti@gmail.com','3 Cantonment','Jhansi','UP'),
('Manoj Tiwari',9888811111,'manoj@gmail.com','6 Market Yard','Nagpur','Maharashtra'),
('Keerthi Kumar',9777711111,'keerthi@gmail.com','18 Temple St','Mysore','Karnataka'),
('Ashok Patil',9666611111,'ashok@gmail.com','77 Shivaji Rd','Kolhapur','Maharashtra'),
('Bhavana Rao',9555511111,'bhavana@gmail.com','14 Residency Rd','Bangalore','Karnataka'),
('Tarun Khanna',9444411111,'tarun@gmail.com','8 Mall Road','Shimla','Himachal'),

('Ritu Malhotra',9333311111,'ritu@gmail.com','6 Garden Rd','Chandigarh','Punjab'),
 ('Sanjay Dutt',9222211111,'sanjay@gmail.com','45 Hill Top','Mumbai','Maharashtra'),
 ('Alok Mishra',9111111112,'alok@gmail.com','5 University Rd','Allahabad','UP'),
 ('Shalini Roy',9000011112,'shalini@gmail.com','22 City Center','Durgapur','WB'),
 ('Farhan Ali',9888811112,'farhan@gmail.com','90 Old City','Hyderabad','Telangana'),
 ('Naveen Gowda',9777711112,'naveen@gmail.com','33 MG Circle','Bangalore','Karnataka'),
 ('Prakash Naidu',9666611112,'prakash@gmail.com','10 Harbour Rd','Vizag','AP'),
 ('Sonia Gill',9555511112,'sonia@gmail.com','17 Golden Temple Rd','Amritsar','Punjab'),
 ('Yogesh Kulkarni',9444411112,'yogesh@gmail.com','9 FC Road','Pune','Maharashtra'),
 ('Chitra Menon',9333311112,'chitra@gmail.com','4 Marine Drive','Kochi','Kerala'),
 ('Ramesh Gupta',9222211112,'ramesh@gmail.com','78 Cloth Market','Indore','MP'),
 ('Dinesh Sharma',9111111113,'dinesh@gmail.com','8 Pink City','Jaipur','Rajasthan'),
 ('Nalini Iyer',9000011113,'nalini@gmail.com','3 Temple View','Chennai','Tamil Nadu'),
 ('Rajesh Khanna',9888811113,'rajesh@gmail.com','6 Carter Rd','Mumbai','Maharashtra'),
 ('Anu Thomas',9777711113,'anu@gmail.com','21 Church Rd','Kottayam','Kerala'),
 ('Satish Reddy',9666611113,'satish@gmail.com','11 Tech Park','Hyderabad','Telangana'),
 ('Vani Krishnan',9555511113,'vani@gmail.com','29 Lake Area','Salem','Tamil Nadu'),
 ('Kishore Das',9444411113,'kishore@gmail.com','7 Hill St','Guwahati','Assam'),
 ('Tejaswini Patil',9333311113,'tejaswini@gmail.com','12 Sugar Mill Rd','Belgaum','Karnataka');

Table 2 vehicles

insert into vehicles(customer_id,vehicle_name,vehicle_type,reg_id,s_id)

values

(1001,'Swift','car',5678,1),
 (1002,'Pulsar','bike',1234,2),
 (1003,'Activa','bike',7788,3),
 (1004,'Innova','car',9988,4),
 (1005,'Eicher','truck',4455,5),
 (1006,'City','car',6677,6),
 (1007,'Splendor','bike',1122,7),
 (1008,'Fortuner','car',3344,8),
 (1009,'Volvo Bus','bus',5566,9),
 (1010,'Mahindra Boat','boat',7789,10),

(1011,'i20','car',2233,11),
(1012,'Apache','bike',4456,12),
(1013,'Tata 407','truck',8899,13),
(1014,'Verna','car',6678,14),
(1015,'Duke','bike',9900,15),
(1016,'Ashok Leyland','truck',4321,16),
(1017,'Jazz','car',7654,17),
(1018,'Access','bike',8765,18),
(1019,'School Bus','bus',3456,19),
(1020,'Fishing Boat','boat',9876,20),
(1021,'Baleno','car',1111,21),
(1022,'Unicorn','bike',2222,22),
(1023,'Safari','car',3333,23),
(1024,'Truck Pro','truck',4444,24),
(1025,'Rapid','car',5555,25),
(1026,'FZ','bike',6666,26),
(1027,'Mini Bus','bus',7777,27),
(1028,'Cargo Boat','boat',8888,28),
(1029,'Alto','car',9999,29),
(1030,'Shine','bike',1212,30),
(1031,'XUV500','car',2323,31),
(1032,'Discover','bike',3434,32),
(1033,'Dump Truck','truck',4545,33),
(1034,'Creta','car',5656,34),
(1035,'Hornet','bike',6767,35),
(1036,'Tour Bus','bus',7878,36),
(1037,'River Boat','boat',8989,37),
(1038,'Ciaz','car',9090,38),
(1039,'Passion','bike',2121,39),
(1040,'Pickup Truck','truck',3232,40),
(1041,'Harrier','car',4343,41),
(1042,'Glamour','bike',5454,42),

(1043,'Luxury Bus','bus',6565,43),
(1044,'Speed Boat','boat',7676,44),
(1045,'Kwid','car',8787,45),
(1046,'R15','bike',9898,46),
(1047,'Trailer Truck','truck',1478,47),
(1048,'Seltos','car',2589,48),
(1049,'Pleasure','bike',3690,49),
(1050,'Metro Bus','bus',1597,50);

Table 3 policy

insert into policy(policy_id,policy_name,premium_amount,duration,coverage_type)

values

(101,'Basic Car',5000,1,'Full'),
(102,'Bike Secure',2500,1,'Third Party'),
(103,'Truck Shield',8000,2,'Full'),
(104,'Boat Care',6000,1,'Partial'),
(105,'Bus Cover',9000,2,'Full'),
(106,'Premium Car',12000,3,'Full'),
(107,'Bike Plus',3500,2,'Full'),
(108,'Truck Heavy',15000,3,'Full'),
(109,'Marine Safe',7000,2,'Partial'),
(110,'Bus Gold',11000,3,'Full'),
(111,'Standard Auto',4500,1,'Partial'),
(112,'Bike Basic',2000,1,'Third Party'),
(113,'Fleet Truck',14000,3,'Full'),
(114,'Boat Basic',5500,1,'Partial'),
(115,'Bus Standard',8500,2,'Full'),
(116,'Car Premium+',13000,3,'Full'),
(117,'Bike Elite',4000,2,'Full'),
(118,'Truck Elite',16000,3,'Full'),
(119,'Boat Elite',7500,2,'Partial'),

(120,'Bus Elite',12000,3,'Full'),
(121,'Economy Car',4200,1,'Third Party'),
(122,'Economy Bike',1800,1,'Third Party'),
(123,'Truck Economy',9000,2,'Partial'),
(124,'Boat Economy',5000,1,'Partial'),
(125,'Bus Economy',7800,2,'Partial'),
(126,'Luxury Car',15000,3,'Full'),
(127,'Sport Bike',5000,2,'Full'),
(128,'Cargo Truck',17000,3,'Full'),
(129,'Luxury Boat',9000,2,'Full'),
(130,'Luxury Bus',14000,3,'Full'),
(131,'Family Car',6500,2,'Full'),
(132,'Daily Bike',3000,2,'Partial'),
(133,'Industrial Truck',18000,3,'Full'),
(134,'Fishing Boat',6200,2,'Partial'),
(135,'School Bus',10000,2,'Full'),
(136,'Car Silver',7000,2,'Full'),
(137,'Bike Silver',3200,2,'Partial'),
(138,'Truck Silver',12500,2,'Full'),
(139,'Boat Silver',6800,2,'Partial'),
(140,'Bus Silver',9500,2,'Full'),
(141,'Car Gold',14000,3,'Full'),
(142,'Bike Gold',4800,3,'Full'),
(143,'Truck Gold',20000,3,'Full'),
(144,'Boat Gold',8500,3,'Full'),
(145,'Bus Gold',15000,3,'Full'),
(146,'Car Platinum',18000,3,'Full'),
(147,'Bike Platinum',6000,3,'Full'),
(148,'Truck Platinum',22000,3,'Full'),
(149,'Boat Platinum',10000,3,'Full'),
(150,'Bus Platinum',17000,3,'Full');

Table 4 insurance

```
insert into insurance(vehicle_id,policy_id,st_date,end_date,ins_status)
```

```
value
```

```
(1001,101,'2023-01-01','2024-01-01','expired'),
(1002,102,'2023-02-10','2024-02-10','expired'),
(1003,103,'2024-01-15','2025-01-15','active'),
(1004,104,'2024-03-01','2025-03-01','active'),
(1005,105,'2023-05-20','2024-05-20','expired'),
(1006,106,'2024-06-10','2025-06-10','active'),
(1007,107,'2023-07-05','2024-07-05','expired'),
(1008,108,'2024-02-01','2025-02-01','active'),
(1009,109,'2023-09-12','2024-09-12','expired'),
(1010,110,'2024-04-18','2025-04-18','active'),
(1011,111,'2023-03-10','2024-03-10','expired'),
(1012,112,'2024-05-05','2025-05-05','active'),
(1013,113,'2023-06-15','2024-06-15','expired'),
(1014,114,'2024-07-20','2025-07-20','active'),
(1015,115,'2023-08-08','2024-08-08','expired'),
(1016,116,'2024-09-01','2025-09-01','active'),
(1017,117,'2023-10-12','2024-10-12','expired'),
(1018,118,'2024-11-11','2025-11-11','active'),
(1019,119,'2023-12-01','2024-12-01','expired'),
(1020,120,'2024-01-25','2025-01-25','active'),
(1021,121,'2023-02-18','2024-02-18','expired'),
(1022,122,'2024-03-09','2025-03-09','active'),
(1023,123,'2023-04-22','2024-04-22','expired'),
(1024,124,'2024-05-30','2025-05-30','active'),
(1025,125,'2023-06-17','2024-06-17','expired'),
(1026,126,'2024-07-19','2025-07-19','active'),
(1027,127,'2023-08-25','2024-08-25','expired'),
(1028,128,'2024-09-14','2025-09-14','active'),
(1029,129,'2023-10-03','2024-10-03','expired'),
```

(1030,130,'2024-11-21','2025-11-21','active'),
(1031,131,'2023-12-15','2024-12-15','expired'),
(1032,132,'2024-01-11','2025-01-11','active'),
(1033,133,'2023-02-07','2024-02-07','expired'),
(1034,134,'2024-03-17','2025-03-17','active'),
(1035,135,'2023-04-29','2024-04-29','expired'),
(1036,136,'2024-05-18','2025-05-18','active'),
(1037,137,'2023-06-09','2024-06-09','expired'),
(1038,138,'2024-07-23','2025-07-23','active'),
(1039,139,'2023-08-30','2024-08-30','expired'),
(1040,140,'2024-09-12','2025-09-12','active'),
(1041,141,'2023-10-18','2024-10-18','expired'),
(1042,142,'2024-11-28','2025-11-28','active'),
(1043,143,'2023-12-24','2024-12-24','expired'),
(1044,144,'2024-01-07','2025-01-07','active'),
(1045,145,'2023-02-20','2024-02-20','expired'),
(1046,146,'2024-03-26','2025-03-26','active'),
(1047,147,'2023-04-11','2024-04-11','expired'),
(1048,148,'2024-05-02','2025-05-02','active'),
(1049,149,'2023-06-13','2024-06-13','expired'),
(1050,150,'2024-07-08','2025-07-08','active');

Table 5 claims

insert into claims (claim_date,claim_amount,claim_status)

values

('2024-01-01',15000,'approved'),
('2024-01-01',18000,'pending'),
('2024-01-02',22000,'rejected'),
('2024-01-02',25000,'approved'),
('2024-01-03',27000,'pending'),
('2024-01-03',30000,'approved'),
('2024-01-04',21000,'rejected'),

('2024-01-04',19500,'approved'),
('2024-01-05',26000,'pending'),
('2024-01-05',32000,'approved'),
('2024-01-06',17500,'pending'),
('2024-01-06',28500,'approved'),
('2024-01-07',23000,'rejected'),
('2024-01-07',31000,'approved'),
('2024-01-08',20000,'pending'),
('2024-01-08',26000,'approved'),
('2024-01-09',28000,'approved'),
('2024-01-09',35000,'pending'),
('2024-01-10',16000,'rejected'),
('2024-01-10',29500,'approved'),
('2024-01-11',22000,'pending'),
('2024-01-11',26000,'approved'),
('2024-01-12',24000,'rejected'),
('2024-01-12',21000,'approved'),
('2024-01-13',33000,'pending'),
('2024-01-13',27500,'approved'),
('2024-01-14',29000,'approved'),
('2024-01-14',31000,'pending'),
('2024-01-15',20500,'rejected'),
('2024-01-15',22500,'approved'),
('2024-01-16',27000,'pending'),
('2024-01-16',18000,'approved'),
('2024-01-17',26000,'approved'),
('2024-01-17',24000,'pending'),
('2024-01-18',32000,'rejected'),
('2024-01-18',35000,'approved'),
('2024-01-19',15000,'pending'),
('2024-01-19',21000,'approved'),
('2024-01-20',19000,'approved'),

('2024-01-20',23000,'pending'),
('2024-01-21',30000,'approved'),
('2024-01-21',25000,'rejected'),
('2024-01-22',27500,'approved'),
('2024-01-22',31000,'pending'),
('2024-01-23',26000,'approved'),
('2024-01-23',28000,'pending'),
('2024-01-24',19500,'approved'),
('2024-01-24',20500,'rejected'),
('2024-01-25',22000,'approved'),
('2024-01-25',24000,'pending');

Table 6 payment

insert into payments(paid_amount, paid_date, payment_mode)

values

(1200,'2025-01-02','cash'),
(850,'2025-01-05','upi'),
(2300,'2025-01-08','card'),
(540,'2025-01-10','cash'),
(1999,'2025-01-12','upi'),
(760,'2025-01-15','card'),
(1450,'2025-01-18','cash'),
(3200,'2025-01-20','upi'),
(410,'2025-01-22','card'),
(2750,'2025-01-25','cash'),
(630,'2025-01-28','upi'),
(980,'2025-02-01','card'),
(1500,'2025-02-03','cash'),
(2100,'2025-02-05','upi'),
(890,'2025-02-08','card'),
(3050,'2025-02-10','cash'),
(450,'2025-02-12','upi'),

(1780,'2025-02-14','card'),
(2600,'2025-02-16','cash'),
(720,'2025-02-18','upi'),
(1100,'2025-02-20','card'),
(900,'2025-02-22','cash'),
(1340,'2025-02-24','upi'),
(2890,'2025-02-26','card'),
(560,'2025-03-01','cash'),
(1700,'2025-03-03','upi'),
(990,'2025-03-05','card'),
(2250,'2025-03-07','cash'),
(680,'2025-03-09','upi'),
(3100,'2025-03-11','card'),
(1250,'2025-03-13','cash'),
(880,'2025-03-15','upi'),
(2050,'2025-03-17','card'),
(770,'2025-03-19','cash'),
(1420,'2025-03-21','upi'),
(3300,'2025-03-23','card'),
(620,'2025-03-25','cash'),
(980,'2025-03-27','upi'),
(1560,'2025-03-29','card'),
(2450,'2025-03-31','cash'),
(510,'2025-04-02','upi'),
(1800,'2025-04-04','card'),
(2950,'2025-04-06','cash'),
(760,'2025-04-08','upi'),
(1180,'2025-04-10','card'),
(2650,'2025-04-12','cash'),
(940,'2025-04-14','upi'),
(2100,'2025-04-16','card'),
(670,'2025-04-18','cash'),

(3200,'2025-04-20','upi');

Table 7 agents

insert into agents(agent_name,phone,branch)

values

('Ravi Kumar',9876543210,'Chennai'),
('Arjun Singh',9123456780,'Mumbai'),
('Vikram Rao',9988776655,'Hyderabad'),
('Karthik S',9090909090,'Chennai'),
('Manoj Sharma',9811122233,'Delhi'),
('Suresh Babu',9345678123,'Bangalore'),
('Ajith Kumar',9567890123,'Chennai'),
('Rahul Verma',9871234560,'Pune'),
('Prakash R',9001122334,'Coimbatore'),
('Dinesh K',9789012345,'Madurai'),
('Sunil Patel',9867543210,'Ahmedabad'),
('Ramesh Gupta',9123987654,'Delhi'),
('Kiran Nair',9898989898,'Kochi'),
('Anand Raj',9445566778,'Chennai'),
('Harish P',9556677889,'Trichy'),
('Deepak Yadav',9876501234,'Lucknow'),
('Gokul Krishna',9345612789,'Chennai'),
('Vivek Jain',9812345678,'Jaipur'),
('Senthil Kumar',9098765432,'Salem'),
('Mahesh B',9786504321,'Bangalore'),
('Naveen Reddy',9988123456,'Hyderabad'),
('Tarun Das',9877098765,'Kolkata'),
('Balaji S',9444400001,'Chennai'),
('Abhishek Mishra',9123009876,'Varanasi'),
('Sathish K',9566001122,'Madurai'),
('Vinod N',9345002233,'Coimbatore'),
('Rohit Sharma',9897001122,'Delhi'),

('Kishore P',9003004455,'Chennai'),
 ('Aravind R',9789005566,'Trichy'),
 ('Nitin Arora',9811006677,'Delhi'),
 ('Surya Prakash',9442207788,'Chennai'),
 ('Ajay Kulkarni',9876508899,'Pune'),
 ('Hari Krishnan',9567809900,'Kochi'),
 ('Muthu S',9345612340,'Salem'),
 ('Ganesh R',9001123456,'Coimbatore'),
 ('Pradeep K',9789015678,'Madurai'),
 ('Lokesh Jain',9812346789,'Jaipur'),
 ('Sanjay Patel',9867544321,'Ahmedabad'),
 ('Varun Gupta',9123988765,'Delhi'),
 ('Rajesh Kumar',9898981234,'Chennai'),
 ('Yogesh B',9556671234,'Bangalore'),
 ('Murali K',9445561234,'Hyderabad'),
 ('Krishna Rao',9988771234,'Vizag'),
 ('Anil Sharma',9811123344,'Delhi'),
 ('Siva Kumar',9090901122,'Chennai'),
 ('Prem Anand',9345672233,'Trichy'),
 ('Karthikeyan',9567893344,'Madurai'),
 ('Ashok R',9001125566,'Coimbatore'),
 ('Subash C',9789017788,'Salem'),
 ('Deepak Raj',9876549988,'Chennai');

Table 8 relations

insert into relations(gender)

values

('M'),('F'),('M'),('F'),('M'),
 ('F'),('M'),('F'),('M'),('F'),
 ('M'),('F'),('M'),('F'),('M'),
 ('F'),('M'),('F'),('M'),('F'),

('M'),('F'),('M'),('F'),('M'),
('F'),('M'),('F'),('M'),('F'),
('M'),('F'),('M'),('F'),('M'),
('F'),('M'),('F'),('M'),('F'),
('M'),('F'),('M'),('F'),('M'),
('F'),('M'),('F'),('M'),('F');

Table 9 services

```
insert into service (service_id,service_date,service_type,service_cost)
values
(1001,'2025-01-02','Oil Change',800),
(1002,'2025-01-04','General Service',1500),
(1003,'2025-01-06','Brake Check',1200),
(1004,'2025-01-08','Full Service',2200),
(1005,'2025-01-10','Engine Service',3000),
(1006,'2025-01-12','Water Wash',600),
(1007,'2025-01-14','Oil Change',900),
(1008,'2025-01-16','Battery Check',1100),
(1009,'2025-01-18','Tyre Service',1400),
(1010,'2025-01-20','Full Service',2600),
(1011,'2025-01-22','General Service',1700),
(1012,'2025-01-24','Brake Check',1300),
(1013,'2025-01-26','Engine Service',2800),
(1014,'2025-01-28','Oil Change',850),
(1015,'2025-01-30','Water Wash',700),
(1016,'2025-02-01','Battery Check',1500),
(1017,'2025-02-03','Tyre Service',1800),
(1018,'2025-02-05','Full Service',3200),
(1019,'2025-02-07','General Service',1600),
(1020,'2025-02-09','Brake Check',1250),
(1021,'2025-02-11','Engine Service',2900),
```

(1022,'2025-02-13','Oil Change',950),
(1023,'2025-02-15','Water Wash',750),
(1024,'2025-02-17','Battery Check',1350),
(1025,'2025-02-19','Tyre Service',1750),
(1026,'2025-02-21','Full Service',3100),
(1027,'2025-02-23','General Service',1650),
(1028,'2025-02-25','Brake Check',1400),
(1029,'2025-02-27','Engine Service',3050),
(1030,'2025-03-01','Oil Change',1000),
(1031,'2025-03-03','Water Wash',800),
(1032,'2025-03-05','Battery Check',1500),
(1033,'2025-03-07','Tyre Service',1900),
(1034,'2025-03-09','Full Service',3300),
(1035,'2025-03-11','General Service',1750),
(1036,'2025-03-13','Brake Check',1450),
(1037,'2025-03-15','Engine Service',2950),
(1038,'2025-03-17','Oil Change',900),
(1039,'2025-03-19','Water Wash',700),
(1040,'2025-03-21','Battery Check',1600),
(1041,'2025-03-23','Tyre Service',2000),
(1042,'2025-03-25','Full Service',3400),
(1043,'2025-03-27','General Service',1800),
(1044,'2025-03-29','Brake Check',1500),
(1045,'2025-03-31','Engine Service',3100),
(1046,'2025-04-02','Oil Change',950),
(1047,'2025-04-04','Water Wash',750),
(1048,'2025-04-06','Battery Check',1550),
(1049,'2025-04-08','Tyre Service',2100),
(1050,'2025-04-10','Full Service',3600);

Table 10 accidents

insert into accident(incident_id, incident_date, location, damage_cost)

values

(1001,'2025-01-02','Chennai',12500.50),
(1002,'2025-01-04','Mumbai',9800.00),
(1003,'2025-01-06','Delhi',15750.75),
(1004,'2025-01-08','Bangalore',8450.25),
(1005,'2025-01-10','Hyderabad',11200.00),
(1006,'2025-01-12','Pune',7600.90),
(1007,'2025-01-14','Kolkata',13400.40),
(1008,'2025-01-16','Ahmedabad',9200.00),
(1009,'2025-01-18','Jaipur',14350.60),
(1010,'2025-01-20','Lucknow',8800.30),
(1011,'2025-01-22','Chennai',15600.00),
(1012,'2025-01-24','Mumbai',9700.45),
(1013,'2025-01-26','Delhi',10100.00),
(1014,'2025-01-28','Bangalore',12200.75),
(1015,'2025-01-30','Hyderabad',14000.00),
(1016,'2025-02-01','Pune',8300.20),
(1017,'2025-02-03','Kolkata',11900.00),
(1018,'2025-02-05','Ahmedabad',9200.90),
(1019,'2025-02-07','Jaipur',11050.00),
(1020,'2025-02-09','Lucknow',7600.10),
(1021,'2025-02-11','Chennai',13450.00),
(1022,'2025-02-13','Mumbai',8900.00),
(1023,'2025-02-15','Delhi',14500.25),
(1024,'2025-02-17','Bangalore',9700.00),
(1025,'2025-02-19','Hyderabad',12800.60),
(1026,'2025-02-21','Pune',8400.00),
(1027,'2025-02-23','Kolkata',13700.30),
(1028,'2025-02-25','Ahmedabad',9200.00),
(1029,'2025-02-27','Jaipur',11100.90),
(1030,'2025-03-01','Lucknow',7800.00),

(1031,'2025-03-03','Chennai',13600.75),
(1032,'2025-03-05','Mumbai',8900.50),
(1033,'2025-03-07','Delhi',15200.00),
(1034,'2025-03-09','Bangalore',9100.00),
(1035,'2025-03-11','Hyderabad',12100.80),
(1036,'2025-03-13','Pune',8400.00),
(1037,'2025-03-15','Kolkata',13300.60),
(1038,'2025-03-17','Ahmedabad',9700.00),
(1039,'2025-03-19','Jaipur',11500.25),
(1040,'2025-03-21','Lucknow',7900.00),
(1041,'2025-03-23','Chennai',14000.00),
(1042,'2025-03-25','Mumbai',9200.30),
(1043,'2025-03-27','Delhi',15000.00),
(1044,'2025-03-29','Bangalore',9800.00),
(1045,'2025-03-31','Hyderabad',12600.40),
(1046,'2025-04-02','Pune',8700.00),
(1047,'2025-04-04','Kolkata',13500.75),
(1048,'2025-04-06','Ahmedabad',9400.00),
(1049,'2025-04-08','Jaipur',11800.00),
(1050,'2025-04-10','Lucknow',8200.00);

select*from accident;

select*from service;

select*from relations;

select * from agents;

select*from claims;

select*from payments;

select *from insurance;

select * from policy;

select*from vehicles;

select * from customers

Customer & Policy Aggregation

List all customers along with:

- Total policy payments made
 - Average policy payment
 - Maximum and minimum policy payment
- Only include customers who have made more than 2 policy payments. Order by total payments descending.

Query

```
select customer_name as name, sum(premium_amount) as  
total_payment_made, avg(premium_amount) as average_payment, max(premium_amount) as  
maximum_payment, min(premium_amount) as minimum_payment from policy inner join customers  
on policy.duration=customers.s_no  
group by customers.customer_name  
having count(policy.duration)>2  
order by total_payment_made;
```

The screenshot shows the MySQL Workbench interface. The SQL Editor contains the following query:

```
614  
615  
616  
617  
618  
619  
620  
621  
622  
-- Average payment  
-- Maximum and minimum payment  
-- Only include customers who have made more than 2 payments. Order by total payments descending.*/  
select customer_name as name, sum(premium_amount) as total_payment_made, avg(premium_amount) as average_payment, max(premium_amount) as maximum_payment, min(premium_amount) as minimum_payment  
from policy inner join customers  
on policy.duration=customers.s_no  
group by customers.customer_name  
having count(policy.duration)>2  
order by total_payment_made;
```

The Results Grid shows the following data:

name	total_payment_made	average_payment	maximum_payment	minimum_payment
Arun Kumar	36500	4055.5556	6000	1800
Priya Sharma	143000	7150.0000	12500	3000
Rahul Das	292300	13919.0476	22000	4800

The bottom panel shows the Action Output with the following messages:

#	Time	Action	Message	Duration / Fetch
30	14:31:08	select * from policy LIMIT 0, 1000	50 row(s) returned	0.000 sec / 0.000 sec
31	14:34:48	select sum(premium_amount) as total_payment_made, avg(premium_amount) as average_payment, max(premium_amount) as maximum_payment, min(premium_amount) as minimum_payment from policy inner join customers on policy.duration=customers.s_no group by customers.customer_name having count(policy.duration)>2 order by total_payment_made;	3 row(s) returned	0.000 sec / 0.000 sec
32	14:35:32	select customer_name as name, sum(premium_amount) as total_payment_made, avg(premium_amount) as average_payment, max(premium_amount) as maximum_payment, min(premium_amount) as minimum_payment from policy inner join customers on policy.duration=customers.s_no group by customers.customer_name having count(policy.duration)>2 order by total_payment_made;	3 row(s) returned	0.000 sec / 0.000 sec

Vehicles & Insurance Status

Show each vehicle's name, customer name, policy name, and insurance status.

Include only vehicles with active insurance.

Sort by customer name alphabetically.

Query:

select vehicle_name, customer_name, policy_name, ins_status as insurance_status from customers
inner join vehicles

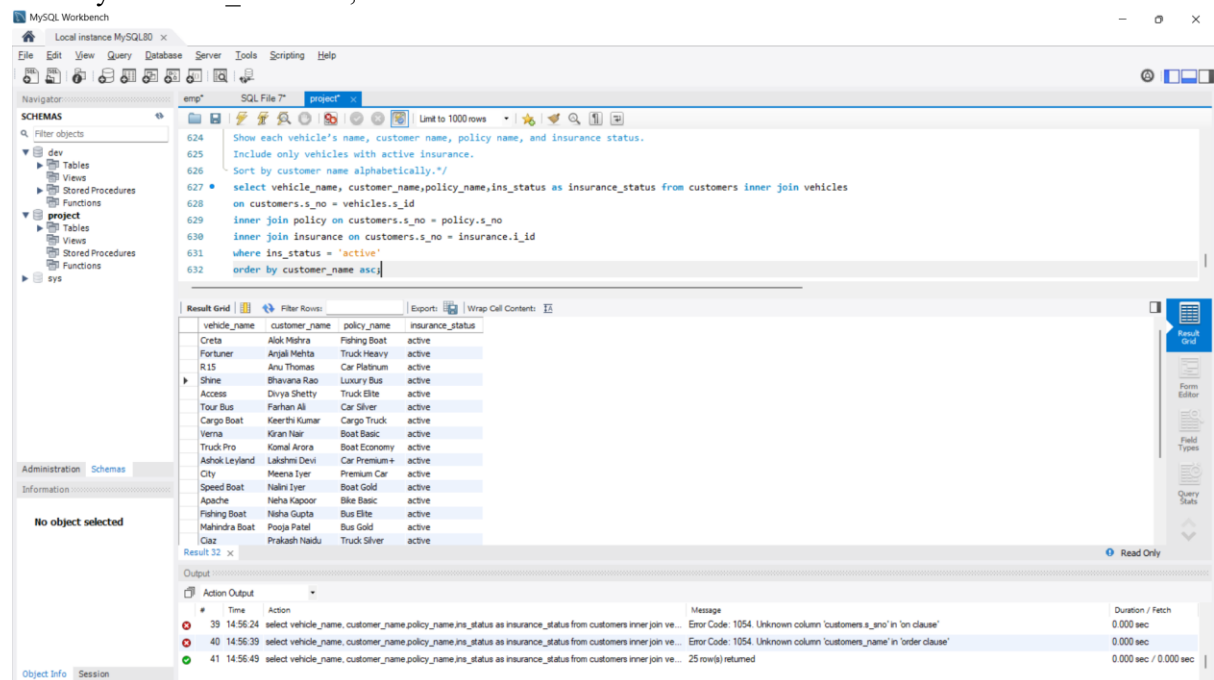
on customers.s_no = vehicles.s_id

inner join policy on customers.s_no = policy.s_no

inner join insurance on customers.s_no = insurance.i_id

where ins_status = 'active'

order by customer_name asc;



Claims Analysis

Find the total number of claims, total claim amount, and average claim amount for each claim status (approved, pending, rejected).

Include only statuses with more than 5 claims.

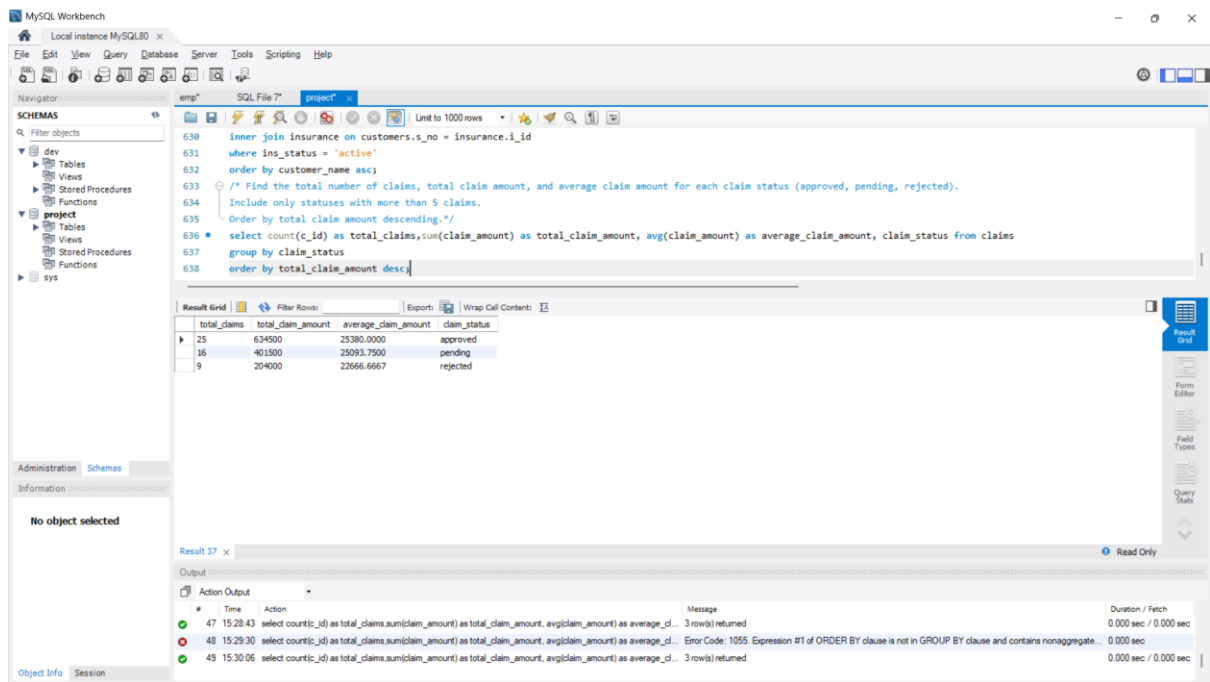
Order by total claim amount descending.

Query:

select count(c_id) as total_claims, sum(claim_amount) as total_claim_amount, avg(claim_amount) as
average_claim_amount, claim_status from claims

group by claim_status

order by total_claim_amount desc;



Service & Accident Join

Get a list of all vehicles that had both a service and an accident.

Show vehicle name, service type, service date, accident date, and damage cost.

Order by accident date descending

Query:

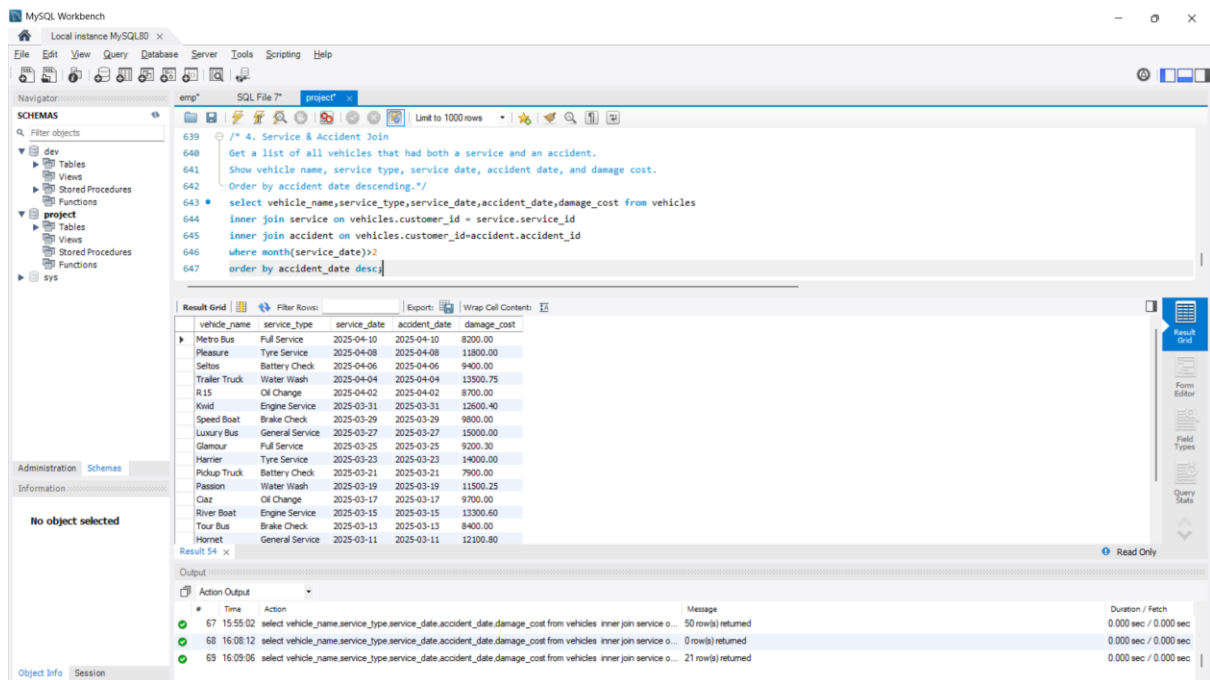
select vehicle_name, service_type, service_date, accident_date, damage_cost from vehicles

inner join service on vehicles.customer_id = service.service_id

inner join accident on vehicles.customer_id=accident.accident_id

where month(service_date)>2

order by accident_date desc;



Policy-Based Vehicle Counts

List each policy name along with:

Number of vehicles using that policy

Total premium collected (sum of premium_amount)

Average premium per vehicle

Include only policies with more than 2 vehicles insured

Query

```

select policy_name,vehicle_name as vehicles_using_that_policy,sum(premium_amount) as
Total_premium_collected,avg(premium_amount) as Average_premium_per_vehicle from policy
inner join vehicles on policy.s_no = vehicles.s_id
where policy.duration > 2
group by policy.policy_name,policy.s_no;

```

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: emp* SQL File 7* project* x

Limit to 1000 rows

649 List each policy name along with:
 650 Number of vehicles using that policy
 651 Total premium collected (sum of premium_amount)
 652 Average premium per vehicle
 653 Include only policies with more than 2 vehicles insured */
 654 select policy_name,vehicle_name as vehicles_using_that_policy,sum(premium_amount) as Total_premium_collected,avg(premium_amount) as Average_premium_per_vehicle from policy
 655 inner join vehicles on policy.s_no = vehicles.s_id
 656 where policy.duration > 2
 657 group by policy.policy_name,policy.s_no;

Result Grid

policy_name	vehicles_using_that_policy	Total_premium_collected	Average_premium_per_vehicle
Premium Car	City	12000	12000.0000
Truck Heavy	Fortuner	15000	15000.0000
Bus Gold	Mahindra Boat	11000	11000.0000
Fleet Truck	Tata 407	14000	14000.0000
Car Premium	Ashtek Leyland	13000	13000.0000
Truck Elite	Access	16000	16000.0000
Bus Elite	Fishing Boat	12000	12000.0000
Luxury Car	FZ	15000	15000.0000
Cargo Truck	Cargo Boat	17000	17000.0000
Luxury Bus	Shine	14000	14000.0000
Industrial Truck	Dump Truck	18000	18000.0000
Car Gold	Harrier	14000	14000.0000
Bike Gold	Glamour	4800	4800.0000
Truck Gold	Luxury Bus	20000	20000.0000
Boat Gold	Speed Boat	8500	8500.0000
Bus Gold	Kwid	15000	15000.0000

Output

Action Output

#	Time	Action	Message	Duration / Fetch
83	10:25:35	select policy_name,vehicle_name as vehicles_using_that_policy,sum(premium_amount) as Total_premium_col...	Error Code: 1055. Expression #2 of SELECT list is not in GROUP BY clause and contains nonaggregated colu...	0.016 sec
84	10:25:37	select policy_name,vehicle_name as vehicles_using_that_policy,sum(premium_amount) as Total_premium_col...	Error Code: 1055. Expression #2 of SELECT list is not in GROUP BY clause and contains nonaggregated colu...	0.000 sec
85	10:25:42	select policy_name,vehicle_name as vehicles_using_that_policy,sum(premium_amount) as Total_premium_col...	21 row(s) returned	0.000 sec / 0.000 sec

Phase 2

Stored procedure

Query

-- stored procedure

-- stored procedure with parameter

delimiter //

create procedure get_claims(in amt int)

begin

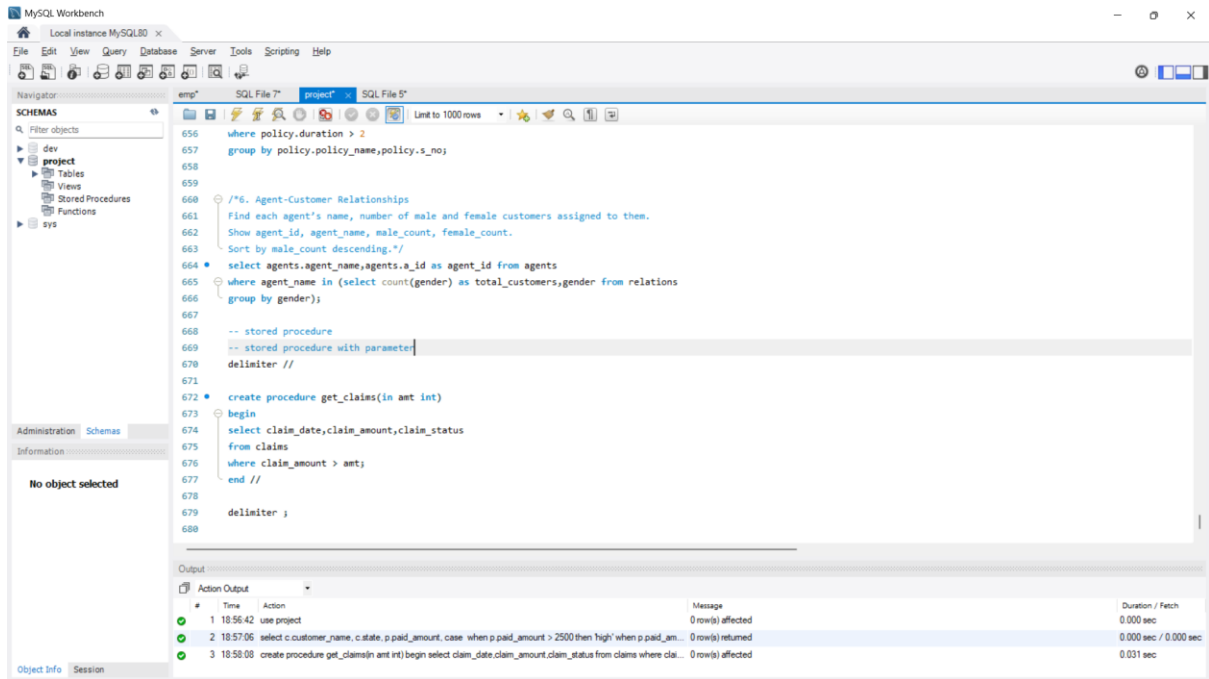
select claim_date,claim_amount,claim_status

from claims

where claim_amount > amt;

end //

delimiter ;



Trigger

Query

-- trigger – before insert

delimiter //

create trigger before_payment_insert

before insert on payments

for each row

begin

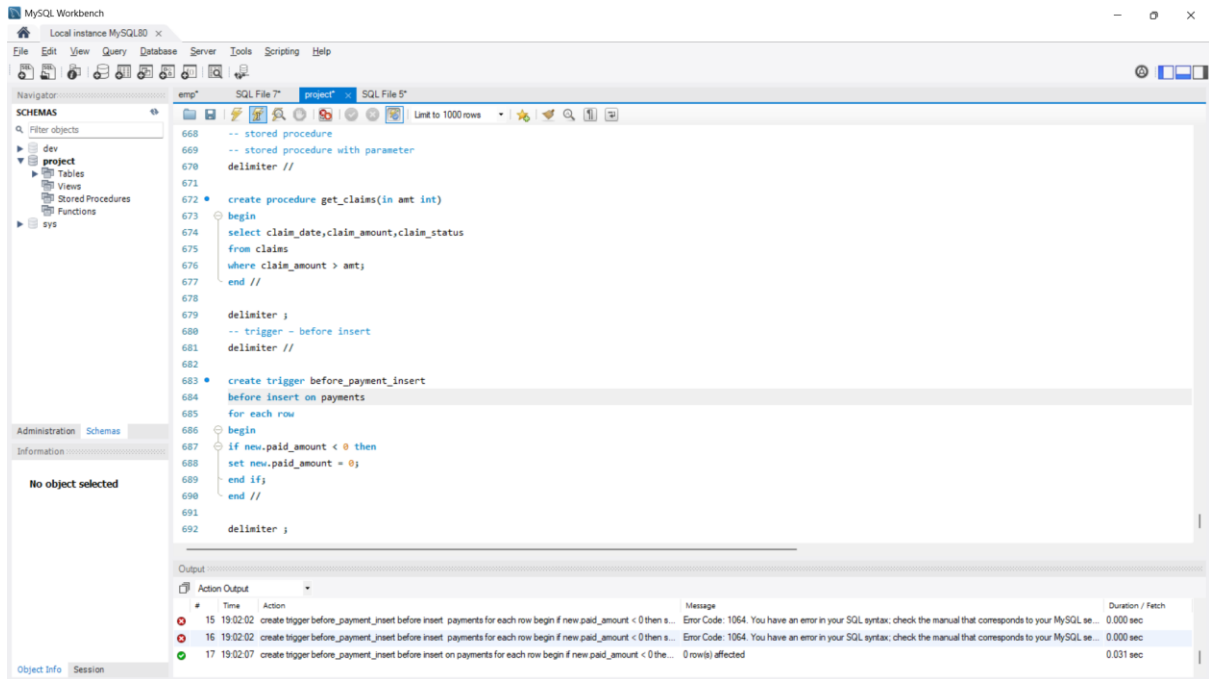
if new.paid_amount < 0 then

set new.paid_amount = 0;

end if;

end //

delimiter ;



window functions

query

-- window functions (rank + dense_rank + row_number)

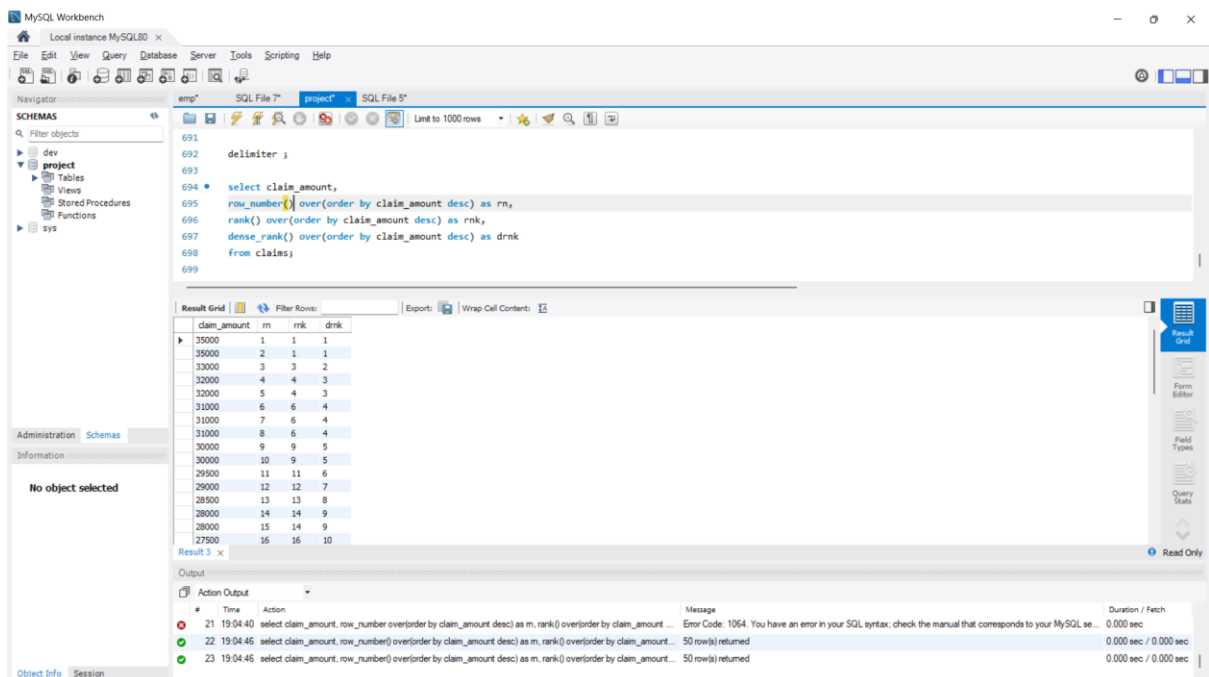
select claim_amount,

row_number() over(order by claim_amount desc) as rn,

rank() over(order by claim_amount desc) as rnk,

dense_rank() over(order by claim_amount desc) as drnk

from claims;



lead & lag (performance analysis)

query

select paid_date,

paid_amount,

lag(paid_amount) over(order by paid_date) as previous_payment,

lead(paid_amount) over(order by paid_date) as next_payment

from payments;

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
698 dense_rank() over(order by claim_amount desc) as drnk
699 from claims;
700
701 -- lead & lag (performance analysis)
702 select paid_date,
703        paid_amount,
704        lag(paid_amount) over(order by paid_date) as previous_payment,
705        lead(paid_amount) over(order by paid_date) as next_payment
706 from payments;
```

The Results tab displays the following data:

paid_date	paid_amount	previous_payment	next_payment
2025-01-02	1200	850	2300
2025-01-05	850	1200	540
2025-01-08	2300	850	1999
2025-01-10	540	2300	760
2025-01-12	1999	540	1450
2025-01-15	760	1999	3200
2025-01-18	1450	760	410
2025-01-20	3200	1450	2750
2025-01-22	410	3200	630
2025-01-25	2750	410	980
2025-01-28	630	2750	1500
2025-02-01	980	630	2100
2025-02-03	1500	980	890
2025-02-05	2100	1500	3050
2025-02-08	890	2100	450
2025-02-10	3050	890	

The Output tab shows the execution log with the following entries:

#	Time	Action	Message	Duration / Fetch
22	19:04:46	select claim_amount, row_number() over(order by claim_amount desc) as m, rank() over(order by claim_amount...	50 row(s) returned	0.000 sec / 0.000 sec
23	19:04:46	select claim_amount, row_number() over(order by claim_amount desc) as m, rank() over(order by claim_amount...	50 row(s) returned	0.000 sec / 0.000 sec
24	19:08:10	select paid_date, paid_amount, lag(paid_amount) over(order by paid_date) as previous_payment, leadpaid_am...	50 row(s) returned	0.016 sec / 0.000 sec